Third Eye: Immediate Evaluation Can Miss Downstream Harm in Verified Language Model Self-Updates

Anonymous ACL submission

Abstract

Verifying a model-generated correction establishes that an example passes a task-specific check. It does not establish the value of training on that example, or the consequences of updating again. We study this distinction in persistent language-model adaptation using three candidate LoRA updates and three native continuations per candidate. A frozen, balanced confirmation comprises 32 fresh initial-state runs across two models and two task families, with independent selection and scoring items. The immediate-selected update has near-zero mean utility relative to holding parameters, but its standardized continuation lowers utility by 1.78 percentage points (simultaneous 95% interval $[-2.88, -0.68]$). The resulting two-update outcome is 1.76 points below holding ($[-3.11, -0.42]$). The predeclared Qwen–Llama difference is 2.80 points ($[0.27, 5.33]$), although a positive-benefit/negative-harm crossover is not established. A development-derived admission rule improves on forced updating but does not outperform holding or protected-case replay. These findings support evaluating the consequences of further adaptation, with uncertainty over both runs and shared evaluation items. They do not identify a mechanism or establish general harm from self-improvement beyond the tested small-batch protocol.

1 Introduction

Language models increasingly contribute to their own adaptation: they generate solutions, identify failures, produce corrections, and supply the data used to change their parameters. Self-generated reasoning can improve task performance (Zelikman et al., 2022; Huang et al., 2022), and self-adaptation can make the production of useful training examples an explicit learning objective (Zweiger et al., 2025). These developments make a seemingly simple decision consequential: should a verified correction become a persistent parameter update?

Three different questions are often adjacent in an adaptation pipeline. Does a correction pass the verifier? Does training on it improve held-out performance immediately? Does the resulting model remain useful after another round of adaptation? A positive answer to the first question does not answer the other two. Moreover, the third question changes the object of evaluation: an updated model generates the data on which its successor is trained. Evaluation of the current checkpoint alone does not measure this subsequent transition.

We introduce *Third Eye*, a controlled measurement framework for these consequences. From a common parent M_0, the framework constructs three equal-budget candidate updates $M_1^{(k)}$. Each candidate then generates verified corrections for three standardized continuation updates, producing $M_2^{(k,d)}$. Candidate selection uses development items; independent scoring items evaluate the parent and every descendant. The no-update alternative is explicit. Figure 1 summarizes the design.

The fresh confirmation yields a clear contrast. The immediate-selected candidate has mean utility $+0.02\,\mathrm{pp}$ relative to the parent, with substantial uncertainty. After the next update, mean utility is $-1.76\,\mathrm{pp}$. The paired continuation contrast is $-1.78\,\mathrm{pp}$, and its multiplicity-adjusted interval excludes zero. This is not evidence that every apparently harmless update conceals damage: an aggregate near-zero mean is not an equivalence or safety test. It is evidence that the immediate endpoint and the continued endpoint give materially different assessments of this update procedure.

The effect also differs across the two tested backbones. Qwen's continued update utility exceeds Llama's by $2.80\,\mathrm{pp}$, meeting the predeclared primary criterion. However, the development pattern "Qwen benefits, Llama is harmed"

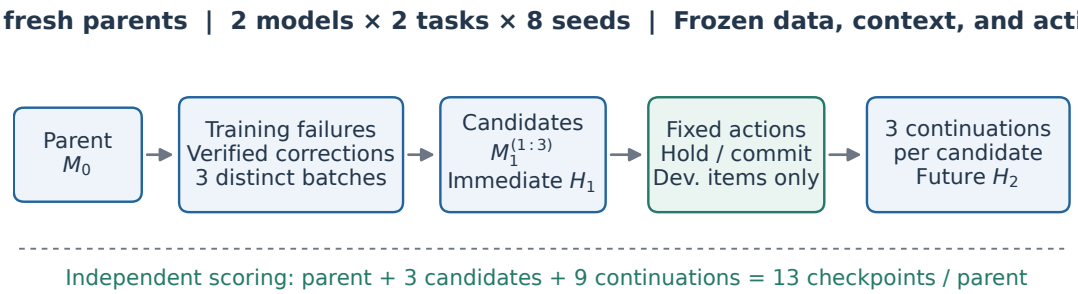


Figure 1: Third Eye separates verified training examples, immediate evaluation, and measured future consequences. All candidate and continuation training budgets are matched. Policy actions are saved before independent scoring begins. Continuations use corrections generated by their own updated model. The parent is scored on the same final items, making holding parameters an explicit baseline.

does not fully replicate: Qwen's pooled mean is slightly negative and its interval crosses zero. Consequently, committing Qwen updates while holding Llama does not demonstrate superiority over holding both.

Our contributions are threefold. First, we provide a prospective, balanced measurement of immediate and continued parameter-update utility with a no-update comparison. Second, we quantify model-dependent outcomes while keeping a positive model contrast separate from evidence of positive adaptation. Third, we provide an auditable evaluation design with repeated continuations, disjoint decision and scoring data, checkpoint verification, and uncertainty over both runs and shared items. Earlier selection and admission experiments provide context, rather than additional independent confirmation. The central contribution is empirical; no forecasting or admission-method advantage is claimed.

2 Controlled Evaluation of Self-Updates

2.1 Corrections, candidates, and continuations

For a model and task family, a parent M_0 attempts training problems. Failed problems receive up to eight correction attempts, and only verifier-passing solutions enter the correction pool. Math uses answer verification; code uses execution in an isolated sandbox. A passing example is evidence about the generated answer under that verifier, not a guarantee about its training effect. This distinction is relevant even with stronger verification (Cobbe et al., 2021; Lightman et al., 2023; Liu et al., 2023).

We sample $K = 3$ distinct candidate batches of two corrections. Problem identities are unique within a batch; batches may overlap across candidates. For each candidate k, the same update operator A gives

$$M_1^{(k)} = A(M_0, B_k; \xi_k). \tag{1}$$

For each of three native draws d, the updated model supplies a fresh correction batch $C_{k,d}$, followed by one equal-budget continuation:

$$M_2^{(k,d)} = A(M_1^{(k)}, C_{k,d}; \zeta_{k,d}). \tag{2}$$

The parent is restored before each candidate, and optimizer state is reset between updates. Thus a candidate does not inherit another candidate's changes or optimizer momentum. Future draws vary the native correction/optimization procedure; they do not independently identify the variance of its constituent sources.

All full updates use LoRA (Hu et al., 2021) with rank 16, scaling 32, learning rate 10^{-4}, and 50 AdamW steps. Candidate data, update counts, and optimizer budgets remain matched. If a continuation cannot supply the required number of verified corrections within its fixed harvesting budget, it preserves $M_2 = M_1$ and records zero continuation steps. No such terminal draw occurred in the fresh confirmation. Appendix A gives the implementation settings and scarcity rules.

2.2 Utility and selection

For role $r \in \{\text{target}, \text{OOD}, \text{retention}\}$, let $a_r(M)$ be accuracy on a fixed item set. We define

utility relative to the parent as

$$U(M; M_0) = \frac{1}{3} \sum_r [a_r(M) - a_r(M_0)]. \quad (3)$$

Reported effects multiply this quantity by 100 and are percentage points, not relative improvements. The equal-role average prevents the larger retention set from receiving a larger weight simply because it has more items. It remains a chosen operational objective, rather than a universal definition of model quality.

The immediate selector chooses the candidate with the largest development utility. Uniform ties are represented as a distribution w_{pk} over candidates for parent run p. All policies use this same candidate distribution; admission changes whether it is committed, not which candidate wins. On independent items, the immediate and future outcomes are

$$G_{1p} = \sum_k w_{pk} U(M_1^{(k)}; M_0), \quad (4)$$

$$G_{2p} = \sum_k w_{pk} \frac{1}{3} \sum_{d=1}^{3} U(M_2^{(k,d)}; M_0). \quad (5)$$

The continuation contrast is the paired quantity $G_{2p} - G_{1p}$. Here H_2 denotes an observed future outcome, not a forecast.

2.3 Frozen policies and independent scoring

We compare four actions. HOLD preserves the parent. FORCED commits the immediate-selected candidate and its standardized continuation. REPLAY commits the same candidate only when its development target gain is positive and both OOD and retention gains are nonnegative. FAMILY is a transparent development-derived baseline: commit Qwen candidates and hold Llama. It is neither a learned predictor nor a proposed general rule for model families. Holding has zero utility by definition on the matched parent comparison; it does not consume an equal training budget.

Policy actions are frozen and durably recorded before final outcomes are read. The final experiment uses one fixed hash-based orientation of the development item split and separate final scoring manifests. Neither candidate choice nor the admission rule is retuned on confirmation outcomes. Each parent contributes 13 scored checkpoints: M_0, three M_1 candidates, and nine M_2 continuations.

Role	Math	Code
Training	GSM8K, 1,024	MBPP, 306
Dev. target	GSM8K, 64	MBPP, 64
Dev. OOD	MATH, 64	MBPP val., 64
Final target	GSM8K, 128	MBPP test, 128
Final OOD	MATH-500, 128	HumanEval, 128
Final retention	MMLU, 256 per task suite	

Table 1: Fixed task suites. Selection and final scoring use separate items. Both development suites include 256 retention items. Code's final OOD domain differs from its development domain; this limits admission-rule transfer.

2.4 Models, data, and experimental units

The confirmation crosses Qwen3-4B (Yang et al., 2025) and Llama-3.2-3B-Instruct with math and code, using eight fresh seeds per stream: 32 initial-state runs in total. The Llama family is described by Grattafiori et al. (2024); the exact checkpoint is the 3.2 instruction model, not a 3.1-8B checkpoint. Parents are repeated runs from a fixed initial model within each stream, not 32 independently pretrained models.

Table 1 lists the suites. GSM8K (Cobbe et al., 2021) and MBPP (Austin et al., 2021) supply training and target problems; MATH (Hendrycks et al., 2021) and HumanEval (Chen et al., 2021) supply the final OOD evaluations. MMLU (Hendrycks et al., 2020) measures retention. Eight HumanEval items used exclusively to measure execution cost are excluded from final scoring. Frozen identifiers and prompt hashes check overlap among the surviving project artifacts. This audit does not establish absence from model pretraining or from unlogged external use.

The confirmation follows a 16-parent development collection and earlier diagnostics. Its seeds, scoring subsets, primary endpoint, and policy actions were declared before fresh outcomes. It contains 96 candidate updates, 288 continuation draws, and 416 checkpoint evaluations. These are repeated measurements within 32 runs, not 288 or 416 independent experimental units. Historical five-generation trajectories remain development evidence and are not pooled into the confirmation.

2.5 Inference and predeclared comparisons

The primary endpoint is mean Qwen-minus-Llama future forced-update utility, macro-averaging math and code. We use 5,000 bootstrap draws, resampling runs within each stream and

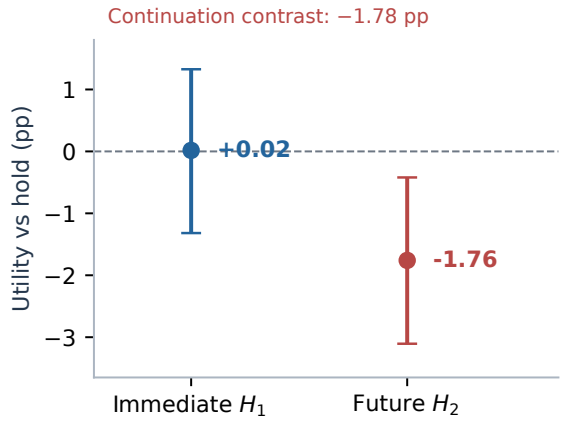


Figure 2: The immediate-selected candidate has near-zero mean utility, whereas its continued outcome is below the parent. Bars show joint run/item 95% intervals. The paired continuation contrast has a separate simultaneous interval $[-2.88, -0.68]$ pp; it is not inferred from overlap between these two bars.

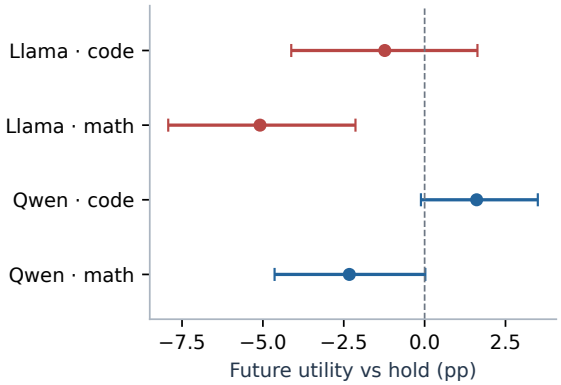


Figure 3: Descriptive future outcomes by model and task, with eight runs per stream and joint 95% intervals. Qwen code's positive mean does not generalize to Qwen math. Stream intervals are not simultaneous confirmatory tests.

evaluation items within each role. Runs using the same exact item pool share an item resample. Separate run-only and item-only analyses are reported alongside the joint analysis; they are not additive causal variance components. This follows the broader need to account for experimental variation and test-set uncertainty (Bouthillier et al., 2021; Dror et al., 2018; Berg-Kirkpatrick et al., 2012).

The six secondary contrasts are FAMILY against hold, forced, and replay; replay against hold and forced; and forced future against forced immediate. We report Holm-adjusted centered-bootstrap p-values (Holm, 1979) and Bonferroni simultaneous intervals for this family. Ordinary policy intervals are explicitly distinguished from those adjusted contrasts. The pooled harm estimate is a policy summary, not a substituted primary endpoint.

A prior joint pooled-harm/model-difference planning criterion did not yield an affordable powered design. The narrower single-primary design was separately declared before confirmation. Its conditional power projection was 85.65%, using 75% of the pilot effect and a 1.25-fold SD inflation. This is conditional planning precision, not post hoc power or a guarantee of successful confirmation.

3 Immediate and Future Utility Give Different Assessments

A near-zero immediate mean precedes a negative future mean. On independent final items, G_1 averages $+0.016$ pp with joint 95% interval $[-1.319, +1.326]$ pp. After the standardized continuation, G_2 averages -1.760 pp with interval $[-3.105, -0.420]$ pp (Figure 2). The paired decline is -1.776 pp, simultaneous interval $[-2.876, -0.679]$ pp, with Holm-adjusted $p = 0.00120$. Thus another equal-budget native update changes the assessment of the procedure.

The paired contrast compares the same selected candidates before and after continuation, scores both on the same independent items, and averages three future draws. The immediate interval nevertheless includes meaningful benefit and harm. A near-zero aggregate mean does not establish equivalence to holding, immediate safety, or individual positive-to-negative sign reversals.

The model difference replicates, but the benefit/harm crossover does not. The primary Qwen-minus-Llama difference is $+2.801$ pp, joint 95% interval $[+0.266, +5.329]$ pp. Model-averaged forced future utility is -3.160 pp for Llama and -0.359 pp for Qwen. Their simultaneous model-mean intervals are $[-5.527, -0.784]$ pp and $[-2.007, +1.439]$ pp, respectively. The positive contrast therefore reflects different update outcomes; it does not establish a beneficial Qwen update regime.

Figure 3 shows why the pooled contrast needs this qualification. Forced future means are -1.23 pp for Llama code, -5.09 pp for Llama math, $+1.61$ pp for Qwen code, and -2.33 pp for Qwen math. In development, both Qwen streams had positive means; that pattern does not persist in

the fresh sample. Model identity alone is consequently an insufficient basis for a general commit decision.

Item uncertainty materially affects precision. For the primary model difference, the run-only interval is $[1.85, 3.78]$ pp, whereas the joint interval is $[0.27, 5.33]$ pp. For forced utility, the corresponding intervals are $[-2.26, -1.27]$ pp and $[-3.11, -0.42]$ pp. The difference matters because many descendants are scored on common items: additional checkpoint evaluations do not make those items independent. Neither interval is a transferable noise floor for other models, tasks, or adaptation systems.

4 What the Evidence Explains—and What It Leaves Open

The continuation result establishes an outcome difference under a fixed procedure. It does not identify whether that difference arises from the candidate's parameter state, the corrections it subsequently generates, optimization on those corrections, or their interaction. Native continuations couple these sources. Treating their variation as separately identified "parameter fragility" and "data-harvest shift" would overstate the design.

An earlier corrected crossed experiment helps bound the interpretation. On eight inspected development parents, it evaluated 216 recipient–donor–draw outcomes, including native and exchanged continuation corrections. The declared native-pairing surplus was -0.19 pp, interval $[-0.61, +0.16]$ pp; the paired test did not establish a native-pairing benefit. A common-donor selector likewise did not demonstrate improvement over immediate selection. These findings leave a specific endogenous-pairing explanation unresolved. They do not show that donor content is irrelevant or that heterogeneous interactions are absent.

A separate eight-parent intervention halved the candidate learning rate while holding continuation learning rate and the remaining protocol fixed. The predeclared selected-candidate effect was -0.24 pp, $[-0.97, +0.37]$ pp, and did not establish a benefit. This is an inconclusive intervention, not proof that smaller updates cannot help. Moreover, reducing learning rate does not reproduce reducing the number of optimizer steps or increasing the amount of unique training data.

The two-example, 50-step recipe is an important boundary of the finding. Repeated optimization on a small verified batch may contribute to the loss; it is not neutralized by the immediate mean being near zero. An additional update itself can be harmful even if the first is not. We therefore describe a measured continuation penalty, rather than a demonstrated latent damage mechanism in M_1. Early stopping and pooled-data updates were not part of the completed fresh confirmation.

Forgetting and update interference are established phenomena (Kirkpatrick et al., 2016; Lopez-Paz and Ranzato, 2017; Zenke et al., 2017; Luo et al., 2023). Fine-tuning can also change out-of-distribution behavior (Kumar et al., 2022), and low-rank adaptation need not preserve the same learned structure as full fine-tuning (Shuttleworth et al., 2024). These are plausible contexts for the observations, not explanations established by this experiment. In particular, aggregate utility loss should not be rewritten as demonstrated retention loss: forced retention is -0.177 pp with interval $[-1.422, +1.139]$ pp, which includes zero.

5 Selection and Admission Require Their Own Evidence

5.1 Selection can inflate the apparent value of candidates

Selecting the maximum of noisy measurements creates optimism even when the underlying candidates are equally useful (Cawley and Talbot, 2010). Our earlier corrected eight-parent analysis measured this effect using all subsets of $K = 1, 2, 3$ candidates, averaging subsets and item-fold orientations within each parent. The same-draw selection optimism was 0.00, 1.25, and 2.09 pp, respectively. Cross-draw gaps were 0.00, 1.20, and 2.25 pp. Figure 4 reports the conditional parent-resampling intervals.

The same-draw analysis changes which items select and score a candidate; the cross-draw analysis also changes the future draw. Their values must not be added as independent variance components. Synthetic equal-value controls exhibited a positive same-outcome maximum but near-zero independently scored selection value. Recoverable-signal controls recovered positive value. These checks validate the analysis behavior in constructed settings, without proving that the real forecasting problem was adequately powered.

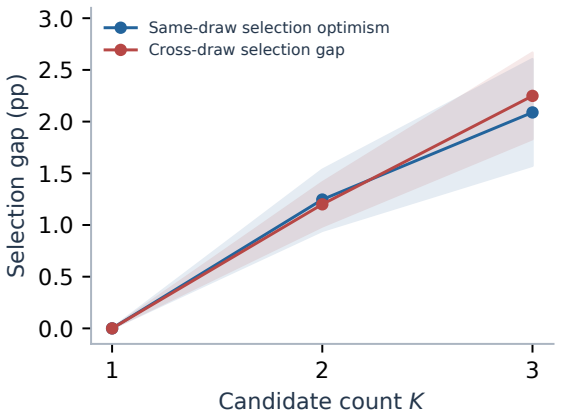


Figure 4: Exploratory selection gaps in the corrected eight-parent development cohort. All candidate subsets are averaged within parents. Bands condition on the finite item pool; this is not fresh confirmation or a universal noise floor. Same-draw optimism and cross-draw disagreement measure different effects.

Policy	Future ΔU	Joint 95% interval	Commit
HOLD	0.00	Reference	0%
FORCED	-1.76	$[-3.11, -0.42]$	100%
REPLAY	-0.27	$[-0.80, +0.21]$	21.9%
FAMILY	-0.18	$[-0.92, +0.60]$	50%

Table 2: Fresh confirmation policy outcomes versus holding, in percentage points. Commit is the fraction of immediate-selected updates admitted. Ordinary joint intervals describe each policy; adjusted pairwise tests are separate.

5.2 Avoiding forced updates is not enough to establish a useful policy

Table 2 compares the four frozen policies. FAMILY improves on FORCED by 1.58 pp, simultaneous interval $[0.14, 2.96]$ pp, Holm $p = 0.0140$. However, it does not outperform HOLD or REPLAY, and its one-sided retention lower bound is -1.06 pp, below the declared -0.5 pp noninferiority margin. It therefore fails the predeclared practical advantage criterion. This distinction prevents an admission policy from being credited for merely avoiding some of a harmful forced-update baseline.

REPLAY improves on FORCED by 1.49 pp nominally, but the simultaneous interval crosses zero and Holm $p = 0.0576$. Neither this comparison nor the near-zero replay-versus-hold mean establishes equivalence or calibrated safety. The correct interpretation is limited evidence for a practical advantage of either inexpensive rule over retaining the initial parameters.

5.3 Forecasting results remain secondary negative evidence

Earlier forecasting experiments did not establish an advantage for learned future-value selection over its immediate-value counterpart. In a balanced historical confirmation, the difference was approximately 0.00 pp, interval $[-0.60, +0.63]$ pp. Historical rendering limitations prevent this collection from serving as clean confirmation of the present phenomenon.

The fresh 16-parent development study is more directly relevant. Immediate-label and future-label admission gates achieved $+0.16$ pp and $+0.13$ pp against holding; their descriptive intervals included zero, and neither established superiority over protected replay. These are finite-data failures to establish advantage, not evidence that future-value prediction is impossible. Forecasting is a distinct supervised task (Luo et al., 2026; Liu et al., 2026); the final 32-parent cohort fits no predictor and makes no architecture claim.

6 Related Work

Learning from model-generated experience. STaR and language-model self-improvement use generated reasoning to construct training data (Zelikman et al., 2022; Huang et al., 2022); self-rewarding models extend the use of model feedback to iterative preference learning (Yuan et al., 2024). SEAL learns to produce self-edits that lead to useful persistent parameter changes (Zweiger et al., 2025). Third Eye studies the evaluation of small verified correction updates and a subsequent native update, rather than introducing a stronger self-edit generator. Output-level refinement (Madaan et al., 2023), verbal reinforcement (Shinn et al., 2023), and the limits of unaided reasoning correction (Huang et al., 2023) concern different objects: an improved response or memory need not be a beneficial weight update.

Self-modification, admission, and prediction. The Darwin Gödel Machine evaluates modifications to an agent's code (Zhang et al., 2025). Rehearse uses prior experimental outcomes in proposal evaluation (Ji and Ding, 2026), and improvement-dynamics prediction has been studied for self-evolving agents (Xie et al., 2026). The Self-Healing Harness governs persistent instruction changes through triggering and protected cases while keeping model weights fixed (Tayebati

et al., 2026). Admission control and local improvement with collateral regression are therefore not new concepts. Our empirical distinction is parameter adaptation followed by another native adaptation step, with independent future scoring and a hold comparison. TuneAhead (Luo et al., 2026) and cross-family micro-interventions (Liu et al., 2026) predict training response; our negative forecasting results do not contradict their findings in different data and training regimes.

Continual adaptation and parameter-efficient updates. Continual-learning methods explicitly address interference across updates (Kirkpatrick et al., 2016; Lopez-Paz and Ranzato, 2017; Zenke et al., 2017), and forgetting during language-model fine-tuning is empirically established (Luo et al., 2023). LoRA and QLoRA provide efficient adaptation mechanisms (Hu et al., 2021; Dettmers et al., 2023), while analyses of low-rank and full fine-tuning reveal structural differences (Shuttleworth et al., 2024). Feature distortion can also affect transfer (Kumar et al., 2022). We do not claim a new forgetting mechanism: the contribution is the measured immediate–continued contrast within a verified self-update procedure.

Verification and reliable evaluation. Outcome and process verification are central to mathematical reasoning (Cobbe et al., 2021; Lightman et al., 2023); code benchmarks and stronger tests expose limitations of apparent program correctness (Chen et al., 2021; Austin et al., 2021; Liu et al., 2023). Our utility combines these task-specific outcomes with broader mathematical and multitask evaluation (Hendrycks et al., 2021, 2020). Work on evaluation reproducibility (Melis et al., 2017), benchmark variation (Bouthillier et al., 2021), significance testing (Dror et al., 2018; Berg-Kirkpatrick et al., 2012), and model-selection bias (Cawley and Talbot, 2010) motivates the decision/scoring split and explicit uncertainty. We adapt established tools to the dependence structure of repeated native parameter updates; we do not present bootstrap intervals or selection optimism as new statistical methods.

7 Conclusion

Under a fixed small-batch LoRA protocol, the immediate-selected update has near-zero mean utility, yet another standardized self-update leads to a significant paired decline and a negative future outcome relative to holding. A predeclared model difference replicates, but positive adaptation for Qwen and a practically superior admission rule do not. The resulting lesson is procedural: verification, immediate utility, and continued utility are separate claims requiring separate evidence. Studies of persistent self-adaptation should include a hold baseline, independently score the effects of further updates, and retain uncertainty over both experimental runs and shared items.

Limitations

The fresh confirmation covers two fixed 3–4B instruction models, math and code, two-example LoRA batches, three candidates, and one standardized continuation. It is not evidence about arbitrary update budgets, larger models, full fine-tuning, or long-horizon autonomous improvement. Initial-state replications do not substitute for prospective five-generation policy trajectories. The 50-step update recipe may be a substantial source of the observed loss; early-stopping and pooled-data comparisons were not performed in this confirmation.

The utility equally weights target, OOD, and retention accuracy. Alternative weights or deployment costs can change a preferred decision. Holding requires no training, while candidate evaluation and disposable training remain real costs even for rejected updates. We do not establish a compute-matched policy advantage. MMLU is one retention proxy, not a comprehensive assessment of capabilities or safety.

Our code OOD evaluation changes from MBPP validation in development to HumanEval in confirmation. This is an intended independent-domain test, but it limits interpretations of admission-rule generalization. Shared benchmark items, finite training pools, verifier limitations, and unknown pretraining exposure also limit population claims. Joint bootstrap inference assumes exchangeable runs within each fixed stream and exchangeable items within the specified pools; it does not represent all possible tasks or pretraining seeds.

The main model difference was predeclared after development, while pooled harm and the continuation contrast have their stated secondary status. The planning projection and final rule condition on finite, inspected development data. Earlier diagnostics and historical trajectories overlap and cannot be summed as independent evidence. A dy-

namic date template affected historical Llama rendering; the fresh confirmation freezes context, but the correction does not retroactively repair every historical claim. Negative forecasting and inconclusive intervention results do not establish impossibility, equivalence, or a causal explanation for the downstream loss.

Ethical Considerations

The study uses public math, code, and multiple-choice benchmarks and existing language-model checkpoints. No human-subject data were collected. Dataset and model licenses continue to govern redistribution; an artifact package should distribute permitted manifests and outcomes without assuming permission to redistribute model weights or all benchmark content. Generated code is evaluated in an isolated execution environment rather than executed with unrestricted host privileges. Accuracy-based retention measures are not safety guarantees. We report negative and inconclusive outcomes alongside positive contrasts to avoid encouraging unsupported deployment decisions. The paper was prepared with AI assistance for drafting, source checking, and figure preparation; the statistical outputs and their provenance remain the basis of its claims.

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A Implementation and Frozen Settings

Setting	Value
Candidate count / batch size	3 / 2
Native continuation draws per candidate	3
Full optimizer steps per update	50
Learning rate / weight decay	10^{-4} / 0
Optimizer	AdamW, reset per update
LoRA rank / scaling / dropout	16 / 32 / 0
Gradient accumulation	8
Maximum sequence length	1,024
Gradient clipping norm	1.0
Maximum generated tokens	512
Correction sampling temperature	0.7
Correction attempts / seed stride	8 / 16
Final scoring per task	128 / 128 / 256
Bootstrap draws	5,000

Table 3: Core frozen settings. The three scoring counts are target, OOD, and retention. Runtime batching is an execution setting; it does not change the unique two-problem update batch or the optimizer-step budget.

LoRA targets attention projection modules. Numeric answers are checked by the benchmark verifier; symbolic math uses an equivalence checker and code uses the isolated test harness. Final scoring is deterministic. Qwen thinking is disabled, and rendering context is frozen, including a constant date of 11 October 2026. Parent and descendant hashes are checked when restoring checkpoints. The complete configuration manifests, rather than model-family defaults, define execution.

Correction scarcity does not authorize shrinking candidate batches, cloning one correction, or admitting unverified examples. Batches are distinct but need not be disjoint across candidates. Every problem identity is unique within a batch. A terminal continuation keeps the exact preceding checkpoint and executes zero optimizer steps; final item outcomes must also preserve equality. All 288 fresh continuations were available, so the present finding does not arise from terminal-state substitution.

B Adjusted Comparisons and Retention

FAMILY was required to beat all three alternatives using simultaneous intervals, exceed the better constant action by at least 0.5 pp, and retain a one-sided 95% lower retention bound above -0.5 pp. These conditions were not met. Inference does not reselect a policy after seeing final scores.

Contrast	Mean	Simultaneous CI	Holm p
FAMILY minus HOLD	-0.180	$[-1.135, +0.862]$	1.0000
FAMILY minus FORCED	$+1.580$	$[+0.143, +2.962]$	0.0140
FAMILY minus REPLAY	$+0.093$	$[-0.629, +0.809]$	1.0000
REPLAY minus HOLD	-0.273	$[-1.039, +0.373]$	0.8458
REPLAY minus FORCED	$+1.487$	$[-0.094, +3.068]$	0.0576
Future minus immediate	-1.776	$[-2.876, -0.679]$	0.0012

Table 4: The complete predeclared secondary family. Simultaneous intervals use Bonferroni quantiles over six contrasts; p-values use Holm adjustment. The procedures differ, so an interval and a test need not give identical decisions in general. Effects are percentage points; positive means favor the left-hand action.

Policy	Retention (pp)	Joint 95% interval
HOLD	0.000	Reference
FORCED	-0.177	$[-1.422, +1.139]$
REPLAY	-0.297	$[-0.724, +0.033]$
FAMILY	-0.515	$[-1.184, +0.122]$

Table 5: Retention is a separate outcome. None of these ordinary intervals establishes negative retention utility. Absence of a detected loss does not establish noninferiority under the declared margin.

C Development Evidence and Separation of Claims

In the 16-parent development cohort, forced utility was -2.31 pp against holding. The immediate-label gate achieved $+0.16$ pp with descriptive parent interval $[-1.01, +1.15]$ pp; the future-label gate achieved $+0.13$ pp with interval $[-0.51, +0.60]$ pp. Their differences from replay also included zero. A privileged cross-draw admission proxy scored $+0.84$ pp against holding. It used future observations to choose admission and other draws/items to score it, so it was neither deployable prediction nor a known upper bound on achievable policy utility.

The corrected recipient–donor experiment's original native-pairing contrast was -0.19 pp, interval $[-0.61, +0.16]$ pp, paired permutation $p = 0.4766$. Native-future selection minus immediate selection was -0.65 pp, $[-1.38, +0.07]$ pp. Common-donor selection minus immediate selection was -0.44 pp, $[-1.68, +0.60]$ pp. These outcomes do not establish a native-pairing explanation for the fresh continuation penalty.

Collection	Size and evidential role
Fresh confirmation	32 runs; 96 candidates; 288 continuations. Frozen primary and policy comparisons.
Fresh development	16 runs; 48 candidates; 144 continuations. Rule development and planning.
Recipient–donor matrix	8 inspected parents; 216 crossed outcomes. Exploratory pairing diagnostic.
Half-LR intervention	8 inspected parents; 24 candidates. Inconclusive; continuation LR unchanged.
Five-generation collections	36 original and 8 balanced trajectories. Historical forecasting context; not pooled.

Table 6: Evidence collections are not additive independent samples. Repeated candidate, draw, and generation counts do not increase the number of independent parent runs. The historical and corrected collections have different conditions.

D Reproducibility, Cost, and Artifact Inventory

The fresh cohort completed all 32 parent jobs and the final analysis without failed parent tasks. Preparation and execution checks preserved original failed cost-estimation attempts rather than silently replacing their records. Full validation passed 209 tests before collection. All training, inference, and scientific statistical analysis ran in scheduler allocations. Login nodes handled staging and scheduler/accounting metadata only.

The final stage used approximately 309.00 node-hour service units under the recorded partition rates and minimum billing, including initial validation and cost-check attempts. These are allocation-based project charges, not energy measurements or a claim about the cost of all historical research. The bounded additional-work ledger totals approximately 542.57 service units. Up to 20 allocated GH200 GPUs ran concurrently; no additional experiment is launched by the final reporting job.

The evidence package contains the frozen declaration, source and scoring manifest identifiers, per-checkpoint item outcomes and score hashes, policy commitments, aggregate results, bootstrap settings, adjusted comparisons, and allocation accounting. The final report SHA256 is `2037f908cd29778d60d1444a12f410fa7d79a86be3783b46768912963a9c1c43`. Hashes and frozen experiment identifiers preserve scientific provenance without requiring access to credentials or computing-account information. Redistribution of model weights is not part of the manuscript package.